

Rama Bala Muppalla

Azure Data Engineer

(682) 306-2447 | ramabalamuppalla16@gmail.com | [LinkedIn](#) |

Professional Summary

Dynamic Azure Data Engineer with 4 years of experience in designing, developing, and managing data warehousing projects across Finance, Sales, and Marketing industries. Expertise in Azure Data Factory (ADF V1, V2) for building and monitoring data pipelines, and proficient in data transformation and automation. Skilled in using Azure Synapse Analytics, SQL Server, and Power BI for creating data models and dashboards. Strong background in ETL processes, data validation, and performance tuning.

Technical Skills

- **Database:** SQL Server 2012/2014, Azure SQL DB, Oracle
- **Big Data:** Azure Data Lake, Cosmos DB
- **Infrastructure as Code (IaC):** Terraform, Terraform Cloud, Terraform Enterprise
- **Reporting:** Power BI, SSRS, Tableau
- **ETL Tools:** Azure Data Factory (ADF), SSIS
- **Analysis Tools:** SSAS, Alteryx
- **Data Modelling Tools:** Erwin, Excel
- **Languages:** SQL, T-SQL, U-SQL, PySpark
- **Tools/Methodologies:** Azure DevOps, GIT, Jenkins, Hive, Erwin, Ansible

Work Experience:

Azure Data Engineer

Jul 2022 – Present

U.S Bank | Minneapolis, MN

- Designed and deployed an Azure-based containerized infrastructure for data processing and analytics, leading to a 20% increase in data processing capacity. Utilized Python for writing data into JSON files for testing Django websites and created scripts for data modeling and import/export.
- Designed and configured databases, backend applications, and programs. Managed large datasets using pandas data frames and SQL. Developed scalable data pipelines for ingesting, translating, and analyzing large sets of data.
- Created job flows using Airflow in Python, automating tasks and scheduling. Managed separate stacks for developing Directed Acyclic Graphs (DAGs) and executed jobs on EMR or EC2 Cluster.
- Built and tested applications, handling database issues and connections with SQL and NoSQL databases like MongoDB. Installed and configured various packages in Python for database operations.
- Utilized Azure Cluster services and Azure Data Factory V2 to ingest diverse data into Azure Data Lake Gen2.
- Implemented Continuous Delivery Pipeline and deployed microservices, provisioning Azure environments, and developing modules using Python and Shell Scripting.
- Imported real-time weblogs using Kafka as a messaging system, ingesting data into Spark Streaming. Conducted data quality checks using Spark Streaming and managed flags for data validation.
- Worked on Big Data Integration & Analytics utilizing technologies such as Hadoop, SOLR, PySpark, Kafka, Storm, and web Methods.
- Responsible for estimating cluster size, monitoring, and troubleshooting of the Spark Databricks cluster. Proficient in applying the Spark Data Frame API for data manipulation within Spark sessions.
- Developed and optimized scalable ETL pipelines using Python, SQL, and Airflow, processing large datasets from AWS Redshift, S3, and RDS.

- Maintained CI/CD pipelines and applied automation to environments and applications using tools like GIT, Terraform, and Ansible.
- Performing Data Validations of the DW using Power BI and Identified data quality issues and design processes and procedures to improve/maintain quality.
- Performance tuning of SQL queries, Data Pipelines, Tableau and Power BI Dashboards.
- Developed and managed cloud infrastructure using Terraform, automating deployments and reducing manual provisioning by 80%.
- Migrated Terraform workspaces from local to Terraform Cloud, improving state management and collaboration.
- Integrated Terraform into CI/CD pipelines using GitHub Actions and Azure DevOps for automated infrastructure provisioning.

Environment: Azure Data Factory, Azure Data Lake, Azure Synapse Analytics (DW), Azure Devops, Snowflake, PowerBI, SharePoint, Windows 10

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Jun 2020 – Dec 2021

- Building ADF pipelines to extract and manipulate data from Azure Blob storage/Azure Data Lake/CosmosDB/SQL Server on cloud.
- Building pipelines using Azure logical apps to extract and manipulate data from Sharepoint.
- Developed scope script using U-SQL to generate daily, monthly structured streams/ORC files in Cosmos.
- Enhancing Multi-Dimensional Revenue Data Cube from ORC files generated in cosmos using TITAN for Business solutions.
- Maintaining and troubleshooting existing Power BI dashboards in case of data refresh failures which are owned by our team.
- Responsible for Windows VM patching on AZURE.
- Performing research to identify source and nature of data required for ETL solutions using Azure Databricks.
- Extensively used Azure Databricks for data validations and analysis on Cosmos structured streams.
- Developing job monitoring alerts for job failures and latency using scope script.
- Extensively used Dragonfly Job Scheduler for scheduling, monitoring the Scope jobs on daily/monthly basis.
- Experience working with Talend ETL and Snowflake cloud database systems.
- Designed and implemented data models (Star and Snowflake schemas) to support analytics workflows, improving query performance by 30%.
- Performing Data Validation between HDInsight's cluster & Target SQL Server using Hive SQL.
- Highly knowledgeable in concepts related to Data Warehouses schemas, Star Schema modeling and Snowflake modeling and processes.
- Developing custom stored procedures for delta loads, functions, triggers using SQL, T-SQL on cloud SQL server.

Environment: Cosmos, Scope Studio, U-SQL, C#, Azure Data Factory, Azure Data Lake, Azure Databricks GIT Hub, Snowflake, Iris Studio, Cauce, Kensho, SharePoint, Windows 10

Education:

Master of Science in Information Studies from Trine University, Detroit, MI, USA.

Bachelors in Computer Science from Vignan's Foundation for Science Technology & Research, India.